MIL-T-47248(MI)

26 July 1974
SUPERSEDING
MIS-13934
4 June 1966

MILITARY SPECIFICATION

TAPE, PLASTIC, HEAT-SEALING, PERFORATED

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a fiber-glass reinforced plastic, perforated, heat-sealing, tear-away tape.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-M-261	Methyl Ethyl Ketone (Technical)
UU-T-450	Tissue, Facial

Military

MIL-A-1154	Adhesive, Bonding, Vulcanized Synthetic Rubber to Steel

FSC8135

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MIL-P-5425	Plastic, Sheet, Acrylic, Heat Resistant
MIL-P-8261	Fungus Resistance Tests, Aeronautical and Associated Materials, General Specification for
MIL-P-14232	Parts, Equipment and Tools for Army Material, Packaging and Packing of
MIL-I-47100	Insulation Sleeving Flexible, Braided, 200 Degrees C (High Temperature Polyamide)

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 1000	Pressure-Sensitive Adhesive Coated Tapes Used for Electrical Insulation, Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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Uniform Classification Committee

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60605.)

National Motor Freight Traffic Associations, Inc.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., 1616 P Street N.W., Washington, D.C. 20036.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample shall meet all the requirements of this specification.

3.2 Material.

3.2.1 Composition and physical properties.

3.2.1.1 Composition. The product shall be a fiberglass-reinforced plastic tape coated with polyamide (nylon) compound, with perforations down the centerline. It shall be capable of heat sealing to itself by application of heat and pressure.

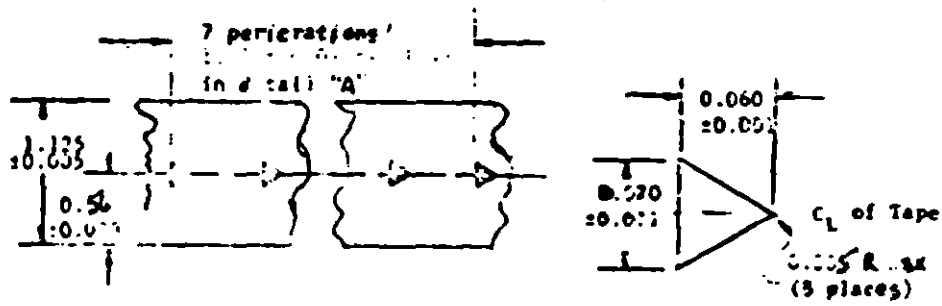
3.2.1.2 Color. The color of the tape shall be black and opaque.

3.2.1.3 Form. Unless otherwise specified (see 6.2), the tape shall be furnished in rolls having a 2-inch core diameter. Splices with overlaps which do not exceed 0.070 inch wide and which provide tear strength in accordance with 3.3.2 are allowed.

3.2.2 Dimensions. The tape width shall be 1.125 inches plus or minus 0.005 inch, and the thickness shall be 0.004 inch plus or minus 0.0003 inch. The thickness requirement does not apply to spliced areas.

3.2.3 Perforations. The tape shall have perforations along its centerline within the dimensional tolerances shown in Figure 1. All material shall be removed from the perforation, except individual cut fibers may cling to edge.

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Detail A perforation

Note: All dimensions in inches.

Figure 1. Tape perforations

3.2.4 Storage life. The tape as furnished by the manufacturer shall be capable of meeting the requirements of this specification after storage in its original package at 60 degrees Fahrenheit (F) to 90 degrees F (16 degrees Celsius (C) to 32 degrees C) for not less than 6 months.

3.3 Performance characteristics.

3.3.1 Heat sealing. The tape shall be capable of forming a strong, uniform bond to itself, and the surface above the bond shall appear glossy.

3.3.2 Tear-away strength. The tape shall have an average tear-away strength along the line of perforations of 1.0 to 3.0 pounds at temperatures of minus 25 degrees plus or minus 5 degrees F (minus 32 degrees plus or minus 3

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degrees C), 77 degrees plus or minus 5 degrees F (25 degrees plus or minus 3 degrees C), and 140 degrees plus or minus 5 degrees F (60 degrees plus or minus 3 degrees C).

3.4 Environmental conditions.

3.4.1 Chemical resistance. The tape shall show no evidence of damage or softening when tested with methyl ethyl ketone (MEK) conforming to TT-M-261 as specified herein.

3.4.2 Fungus resistance. The tape shall support no fungus growth visible to the unaided eye.

3.4.3 Heat resistance. The tape shall exhibit no visible distortion or change in dimensions when heated as specified herein.

3.5 Workmanship. The workmanship shall be such as to enable the tape to be free from flaws, defects, pinholes, bubbles, cracks, or inclusions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. When specified in the contractual document (see 6.2), a sample consisting of 1 roll of 25 feet of tape, manufactured under the same conditions as those proposed for subsequent production, shall be subjected to preproduction inspection. Accomplishment of preproduction inspection shall be as specified herein. If the preproduction sample does not meet the requirements of this specification, it shall be rejected. Subsequent units shall not be considered for acceptance until Government

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approval of the preproduction sample has been obtained. Units subjected to preproduction inspection shall have successfully passed acceptance inspection. Testing of the preproduction sample, to determine compliance with the characteristics listed in Table I, shall be conducted in accordance with the corresponding test paragraphs.

Table I

Preproduction inspection

Characteristic	Requirement source	Test paragraph
Composition and physical properties	3.2.1	4.5.3.1
Dimensions	3.2.2	4.5.3.2
Perforations	3.2.3	4.5.3.3
Storage life	3.2.4	4.5.3.4
Heat sealing	3.3.1	4.5.3.5
Tear-away strength	3.3.2	4.5.3.6
Chemical resistance	3.4.1	4.5.3.7
Fungus resistance	3.4.2	4.5.3.8
Heat resistance	3.4.3	4.5.3.9
Workmanship	3.5	4.5.4

4.3 Quality conformance inspection.

4.3.1 Sampling for acceptance inspection. Sampling for acceptance inspection shall be conducted in accordance with MIL-STD-105, at an acceptable quality level (AQL) of 1.5 percent defective. A roll of tape shall be considered a unit of product. Randomization of the containers in the lot, as well as randomization of the location and number of rolls (or units) within each container, shall be employed when sampling for acceptance inspection.

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4.3.2 Lot formation. A lot shall consist of all the material manufactured in one continuous operation by the same process by the same manufacturer in accordance with this specification and submitted for inspection at one time.

4.3.3 Acceptance inspection. Acceptance inspection of the sample specified in 4.3.1, to determine compliance with the characteristics specified in Table II, shall be conducted in accordance with the corresponding test and inspection paragraphs.

Table II

Acceptance inspection

Characteristics	Requirement source	Test and inspection paragraph
Composition and physical properties	3.2.1	4.5.3.1
Dimensions	3.2.2	4.5.3.2
Perforations	3.2.3	4.5.3.3
Heat sealing	3.3.1	4.5.3.5
Tear-away strength at 77 degrees F	3.3.2	4.5.3.6
Workmanship	3.5	4.5.4
Packaging, packing and marking	5.	4.6

4.4 Inspection equipment. The inspection equipment for conducting examination and tests shall be as specified in the applicable test methods and procedures paragraphs.

4.5 Test methods and procedures.

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4.5.1 Test conditions. Unless otherwise specified herein, the following conditions shall be used as a basis to establish performance requirements.

a. Temperature, room ambient plus or minus 9 degrees C (16 degrees F).

b. Altitude, facility ground.

c. Humidity, facility ambient up to 95 percent relative humidity.

4.5.2 Test sequence. Testing within each classification of inspection (preproduction or acceptance) shall be at the option of the supplier.

4.5.3 Tests.

4.5.3.1 Composition and physical property. Composition shall be certified by the supplier and the physical properties shall be examined visually to establish compliance to 3.2.1.

4.5.3.2 Dimensions. The width shall be determined by the use of a suitable instrument graduated to read within 0.001 inch. The tape shall be flat on a smooth surface at the time of measurement. At least 10 measurements shall be made. The thickness shall be determined in accordance with ASTM D 1000. Dimensions shall conform to 3.2.2.

4.5.3.3 Perforations. A specimen of tape shall be placed between 1/6-inch sheets of heat resistant acrylic plastic conforming to MIL-P-5425. The dimensions and spacing of perforations shall then be measured with a Kodak Optical Comparator, or equivalent, and shall be as specified in 3.2.3.

4.5.3.4 Storage life. One roll of tape in its original package shall be stored under the conditions specified in 3.2.4. At the end of the storage period, the tape shall be subjected to all the tests contained herein. The supplier's certificate of compliance certifying to the storage life requirements (see 3.2.4) may be accepted during the storage period.

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4.5.3.5 Heat sealing. A length of tape at least 6 inches long shall be folded double along its centerline. The doubled area shall be pressed with a heated tool maintained at a surface temperature between 360 degrees and 400 degrees F (182 and 204 degrees C). The time of contact of the tool with the tape shall not exceed 1 second. The temperature shall be adjusted in such a manner as to cause the overlapped layers of tape to bond together without blistering or burning of the tape. A correct bond shall have strength sufficient to prevent pulling the layers apart without tearing the tape and shall conform to 3.3.1.

4.5.3.6 Tear-away strength. A polyamide (nylon) sleeving conforming to MIL-I-47100, of approximately 0.035 inch outside diameter with six (6) 0.005 inch steel wires enclosed, shall be placed along the perforated line on the tape. The tape shall be folded over at the perforation line and heat sealed as specified in 4.5.3.5, with the sleeving encapsulated at the perforation line. The full length of the tape shall then be bonded with adhesive conforming to MIL-A-1154 or clamped by a suitable means to a flat aluminum plate. A length of sleeving which has been left extending out of the tape retainer shall be used to apply a pulling force parallel to the perforation line approximately in a plane with the tape as bonded or clamped to the plate. The angle created between the sleeving being pulled and the perforation line shall be approximately 10 to 15 degrees. The average tear-away force (i.e., force causing the sleeving to tear out of the perforations) shall be measured as the sleeving is being torn away at a rate of 5 to 15 inches per second, at each of 3 temperatures: minus 25 degrees plus or minus 5 degrees F (minus 32 degrees plus or minus 3 degrees C); 77 degrees plus or minus 5 degrees F (25 degrees plus or minus 3 degrees C); and 140 degrees plus or minus 5 degrees F (60 degrees plus or minus 3 degrees C), and shall be as specified in 3.3.2.

4.5.3.7 Chemical resistance. Chemical resistance of the tape shall be tested as specified below and shall conform to 3.4.1.

Place a small pad of tissue conforming to UU-T-450, approximately 1 inch square and 8 to

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12 layers thick, on the tape surface. Fully saturate with methyl ethyl ketone conforming to TT-M-261. Cover the pad with a watchglass or small beaker. Remove the pad after 1 minute and rub the moistened spot with a clean dry cloth or tissue. Failure is indicated by discoloration of the cloth or tissue or by visible evidence of damage to the tape surface. A slight whitening of the surface which may be removed by light rubbing with a dry cloth is not reason for rejection. Resistance to softening is determined by scratching the surface with a fingernail.

4.5.3.8 Fungus resistance. The tape shall be tested for fungus and mildew resistance in accordance with MIL-F-8261, and shall be as specified in 3.4.2.

4.5.3.9 Heat resistance. A short length of tape shall be held horizontally above the flame of a match or alcohol burner in such a manner that the plastic matrix is ignited and burns. After burning ceases, the tape shall be visually inspected for evidence of warping, curling, or change of size. The results shall be as specified in 3.4.3.

4.5.4 Workmanship. The tape shall be visually examined to assure conformance to the workmanship requirement specified in 3.5.

4.6 Packaging, packing and marking. The inspector shall ascertain that the packaging, packing and marking of the material conforms to this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified by the contract or purchase order, preservation, packaging, and packing requirements for the specified levels of protection shall be in accordance with MIL-P-14232, Uniform Freight Classification Rules and National Motor Freight Rules.

5.2 Marking. Marking shall be in accordance with MIL-STD-129. Marking shall include, but not be limited to the following information:

a. Manufacturer's name and address.

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b. Manufacturer's designation.

c. Lot number or batch number.

d. Date of manufacture.

e. Shelf life or storage limitations.

f. This specification number.

6. NOTES

6.1 Intended use. This tape is intended for use as a tear-away fastening device for positioning lead wires on the exterior of missiles. It is suitable for use between minus 25 degrees and 140 degrees F (minus 32 degrees and 60 degrees C) after storage between minus 80 degrees and 155 degrees F (minus 62 degrees and 68 degrees C).

6.2 Ordering data. Procurement data shall specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required (see 3.1), and if so, pertinent details.

c. Form, if different from 3.2.1.3.

d. Requirement for certification of composition (see 4.5.3.1).

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-13934, dated 4 June 1966.

Custodian:
Army-MI

Preparing Activity:
Army-MI
Project No. 8135-A421

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DOCUMENT IDENTIFIER AND TITLE

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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